Gas Exchange and Oxygenation



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Hyperventilation

- Increase in respiratory rate
- Increased in carbon dioxide (CO_2) exhaled
  - Lower levels of CO_2 in blood
- Causes:
  - Anxiety attack
  - Infection such as pneumonia
  - Lung disease
    - COPD
    - Asthma
  - Diabetic ketoacidosis
  - Brain injury
  - Drug use



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During hyperventilation, an increased amount of carbon dioxide is exhaled, leading to lower than expected levels of carbon dioxide in the blood. These low levels of carbon dioxide can cause the client's arterial pH to increase and alkalosis to develop.

Hyperventilation cont.

Manifestations

- Increased respiratory rate (>20)
- Weakness
- Dizziness
- Headache
- Anxiety
- Increased heart rate
- Difficulty breathing
- Numbness/tingling in fingers

Interventions

- Breathing techniques
- Medication
- Airway management

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Hypoventilation

- Decrease in respiratory rate
- Increased in CO2 in blood
- Causes:
  - Neuromuscular disorders
    - Muscular dystrophy
  - Medications
    - Barbiturates
    - Narcotics
    - Benzodiazepines
  - Neurologic disorders
  - Trauma
  - Lung diseases
    - COPD
    - Asthma



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Hypoventilation is shallow breathing with a lower than expected respiratory rate. It leads to too much CO_2 in the blood and not enough oxygen. Elevated CO_2 levels increase the amount of carbonic acid in the blood. This decreases the pH of the blood and results in acidosis.

Hypoventilation cont.

Manifestations

- Decreased respiratory rate (<12)
- Anxiety
- Weakness
- Disturbed sleep
- Change in level of consciousness
- Dysrhythmias

Interventions

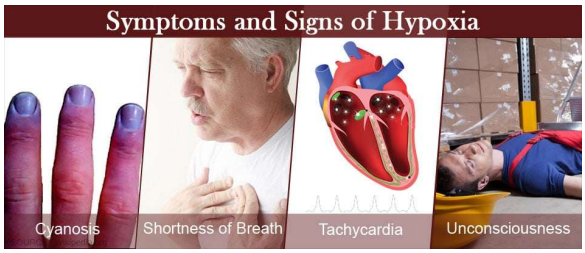
- Breathing techniques
- Medication
- Airway management

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Hypoxia

- Lack of oxygen at cellular level
- Decreased amount of oxygen to tissues
- Causes
  - Low red blood cells
  - Decreased ventilation
  - Poor perfusion
  - Smoke inhalation
  - High altitudes
  - Lung diseases (COPD, pneumonia, asthma)
  - Medications that decrease RR
  - Anemia

Interventions:
- Supplemental oxygen
- Monitor Sp02



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How do you think someone with hypoxia may feel? Look?

Oxygen Delivery Devices

- Considered a medication
- Requires an order
  - Continuous, PRN, titration
- Can help decrease work of breathing
- Sp02 between 95% and 100% helps prevent hypoxemia
  - Different range for clients with chronic illness such as COPD
- Tanks must be secured
- Check levels in tanks



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Supplemental oxygen may be required for clients with cardiopulmonary issues to achieve a fraction of inspired oxygen greater than 21%—the percentage of oxygen in room air. This therapy is used to treat hypoxemia.

Nasal Cannula

- Deliver oxygen via prongs in nares
- Used for low flow 1L to 6L/min
- Monitor for skin breakdown
- Can add humidification
  - Oxygen mixed with water
  - Prevent dry mucous membranes
- High-flow nasal cannula
  - Non-invasive alternate
  - Can provide higher flow rates





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Simple Face & Partial Rebreather Mask

- Simple Face Mask
  - Deliver rates of 5L to 10L/min
  - Retains CO2 when client breathes
  - Monitor for skin breakdown



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Partial Rebreather Mask

- Partial Rebreather Mask
  - Similar to simple mask but has a reservoir bag
  - One way valve – allows for small amount of outside air to be breathed in
  - Deliver rates of 10L to 15L/min



- Fill reservoir bag with oxygen
- Oxygen draws into mask from inflated bag, so oxygen/CO2 mixture is diluted with high-flow oxygen
- Reservoir bag must remain inflated to prevent retention of CO2 that is exhaled

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Nonrebreather

- Nonrebreather Mask
  - Similar to partial rebreather with reservoir bag
  - One-way valve ensures no outside air breathed
  - Ensure snug fit
  - Delivers 10L to 15L/min
  - Contraindicated for client with COPD
  - Caution with long-term use during respiratory failure



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Venturi Mask

- Venturi Mask
  - Fixed concentration device
  - Delivers precise amount
  - Rates of 4L to 15L/min
  - Change barrels to change amount of oxygen

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CPAP

CPAP

- Continuous positive airway pressure
- Delivers flow through mask or nosepiece
- Constant, steady flow to keep alveoli open and improve oxygen in blood
- Supplemental oxygen can be added to system
- Used primarily for sleep apnea

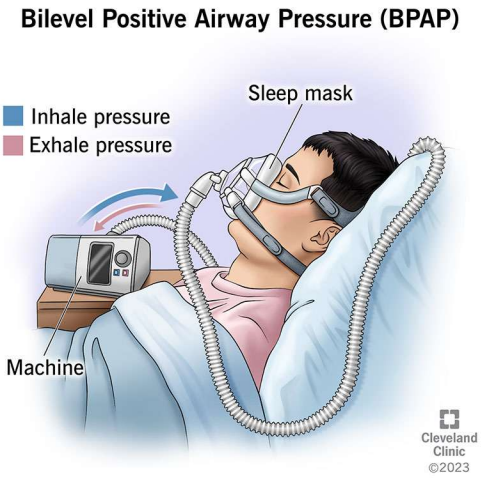


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BiPAP

BiPAP

- Bilevel positive airway pressure
- Higher pressures when inhaling
- Lower pressure when exhaling
- Can use if airway collapses during sleep
- Used for clients with disorders that cause muscle weakness inhibiting breathing
  - COPD
  - Heart failure



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Discharging with Oxygen

- Walk Study
  - 6 minute walk test
  - Respiratory therapist (RT) walks with client in hallway monitoring SpO2 levels
- RT recommends oxygen delivery system and educates client on the equipment for home
- Case management identifies oxygen vendor and coordinates delivery
  - Ensures equipment is delivered prior to discharge



- Nurse role:
  - Educate:
    - Oxygen safety
      - No open flames
      - No smoking
      - No petroleum-based products near oxygen
      - Keep away from heat sources
    - Equipment care
    - Correct liter flow settings and limits
    - When to call the provider
  - Document education and any return demonstration

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Home Safety with Oxygen

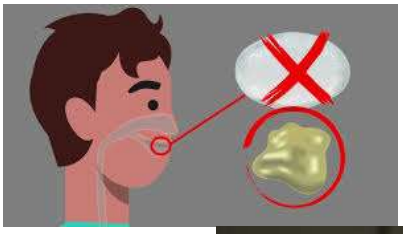
- Combustible under certain conditions
- Education:
  - Never smoke around oxygen
  - Keep away from open flames
  - Keep oxygen containers upright and secured
  - Keep fire extinguisher within home



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Interventions to Improve Oxygenation

- Sputum Collection
  - Assists in diagnosis and treatment
  - Collect before eating or drinking
  - Try to collect 1-2tsp in sterile cup
  - May need chest physiotherapy to loosen secretions
- Chest Physiotherapy
  - Percussion of chest with vibration
  - Helps loosen thick secretions
  - Lasts 20-30 minutes
- Incentive Spirometer
  - Used to promote deep breathing
  - Provides feedback on inspiratory volume
  - Educate on proper use
  - Help prevent atelectasis







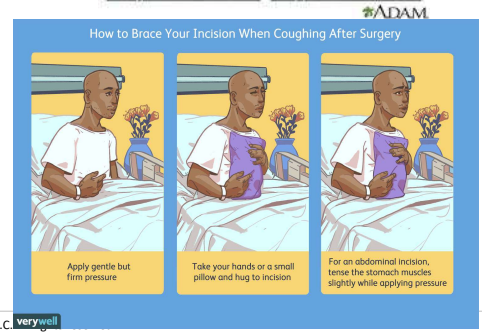
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Nurses should educate clients on the purpose of the IS and proper use of the device, and should ensure that the clients perform a return demonstration. When using the IS, clients should understand that the device is to be used 10 repetitions per hour with each breath held for 3 to 5 seconds. Nurses should assist clients in setting a target goal and explain the importance of using the IS. With continued use, pulmonary function will improve and breath sounds will become clear.

Interventions to Improve Oxygenation cont.

- Purse-lipped breathing
  - Technique to help with breathlessness
  - Can help release air trapped
  - Promotes CO2 elimination
  - Relax neck and shoulders and breath in through nostrils and out through pursed lips
- Coughing & deep breathing
  - Prevent atelectasis and pneumonia
  - Encourage regular use
  - Use pillow to splint abdomen and chest



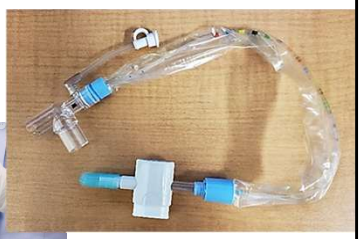


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Tracheostomy

- Airway management device
- Special equipment for suctioning required
  - Open vs closed system
  - Monitor Sp02 during procedure
- Care of trach needed to maintain patency
  - Cleanse stoma site and plate
  - Change gauze under plate
  - Replace inner cannula
  - Change ties each day



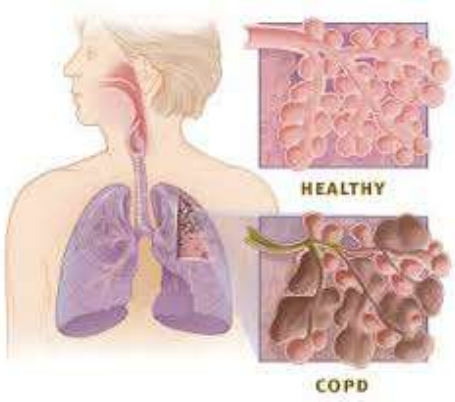




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COPD and COPD Exacerbation

- Characterized by persistent respiratory symptoms and airflow limitations
- Pathologic changes in airways
- Risk factors:
  - Tobacco use
  - Environmental and occupational exposure to dust/fumes
  - Respiratory infections
  - Extreme temperatures
  - Treatment nonadherence



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COPD and COPD Exacerbation cont.

- Manifestations:
  - Dyspnea
  - Coughing
  - Wheezing
  - Fatigue
  - Changes in mucous
  - Swelling of legs or ankles
  - Low Sp02

- Interventions:
  - May need to increase oxygen
    - Never above 6L/min
    - May require BiPAP
  - Breathing treatments
  - Steroids
  - Mucolytics
  - Pursed lip breathing and tripoding



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Questions??



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